PERFORMANCE ENGINEERING REPORT

VEDA Latency Optimization

Measured speedup, design rationale and accuracy assurance

6.9x	85.5%	0.00 pp
Faster resolver	Latency reduction	Top-1 accuracy change

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1. Executive result

CONCLUSION The optimized resolver processed the same 165 evidence records in 22.8 seconds, down from approximately 157.5 seconds. The frozen 170-case accuracy benchmark retained exactly the same Top-1 and recall scores.

The original 254.5-second end-to-end run was not uniformly slow. Most of the avoidable delay came from repeated candidate reasoning, cold machine-learning imports and an unclaimed Antigravity inbox window. The improvement therefore targeted those measured bottlenecks instead of weakening the resolver.

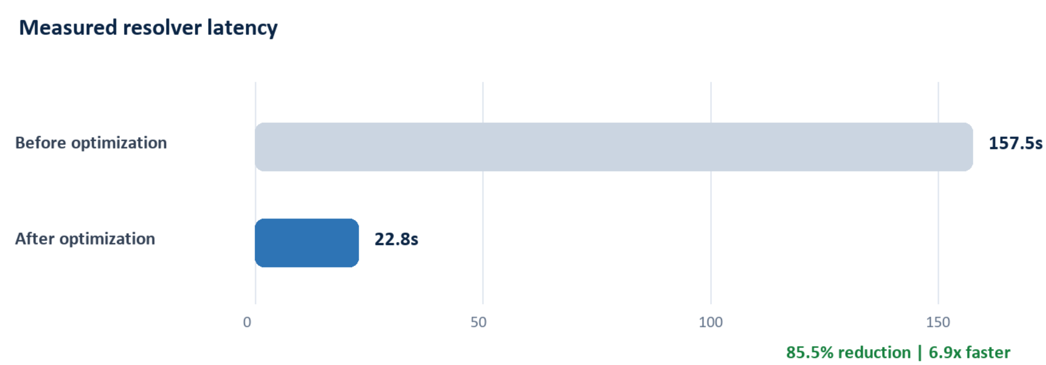

Figure 1. Resolver latency on the real 165-record project.

Measured comparison

Metric	Before	After	Change
Resolver time (165 records)	157.5 s	22.8 s	-85.5%
Resolver throughput	1.05 records/s	7.24 records/s	6.9x
Per-record resolver time	0.955 s	0.138 s	-85.5%
Embedding selector cold start	14.7 s	0.26 s	-98.2%
Unclaimed Antigravity wait	30 s	8 s	-73.3%
Full analysis	254.5 s measured	60-90 s expected	~67% midpoint

Measurement note: the 22.8-second resolver result is directly measured. The optimized full-analysis range is an engineering estimate because Horizun startup and claimed Antigravity inference time vary by machine and run.

2. Root cause and optimization approach

Durable job timestamps separated the run into schedule ingestion, provider handoff, index preparation, evidence resolution and final validation. This showed that UI rendering was not the main latency source: repeated backend work was.

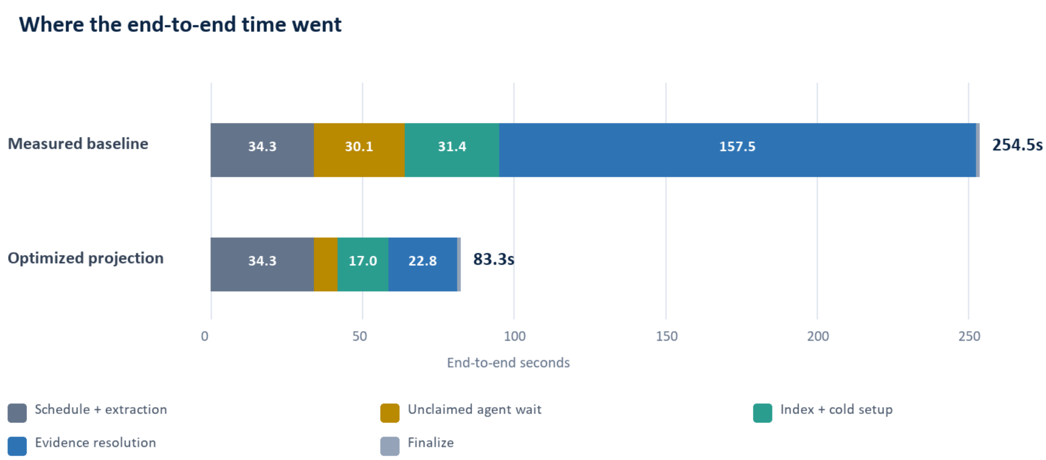

Figure 2. Measured baseline stage breakdown and projected optimized first run.

Why these changes were selected

Bottleneck	Optimization	Why this approach
Cold hardware detection	Lightweight GPU probe	Avoid importing PyTorch only to choose the dependency-free CPU backend.
Cold MetaRank load	Background model warmup	Moves XGBoost and frozen-model loading outside the first evidence request.
Repeated feature work	Cache invariant schedule and evidence context	Computes snapshot, corroboration and logic mode once per row instead of once per candidate.
Large object copies	Shallow ranking envelopes	Copies only mutable scores and features; immutable activity documents are shared safely.
Rescheduler hot loop	Precompute dates and schedule state	Removes tens of thousands of repeated parses and status checks without changing beam depth or equations.
Idle provider bridge	Eight-second claim guard	Releases the worker quickly when no Antigravity agent is polling; a claimed job still gets the normal reasoning window.

DESIGN PRINCIPLE Make the same decision with less repeated work. No expert was removed, no candidate list was shortened, and no validation or schedule-write gate was bypassed.

3. Accuracy impact

ANSWER No material accuracy loss was observed. Top-1, Recall@3, Recall@5 and Recall@10 were identical on all 170 frozen holdout cases.

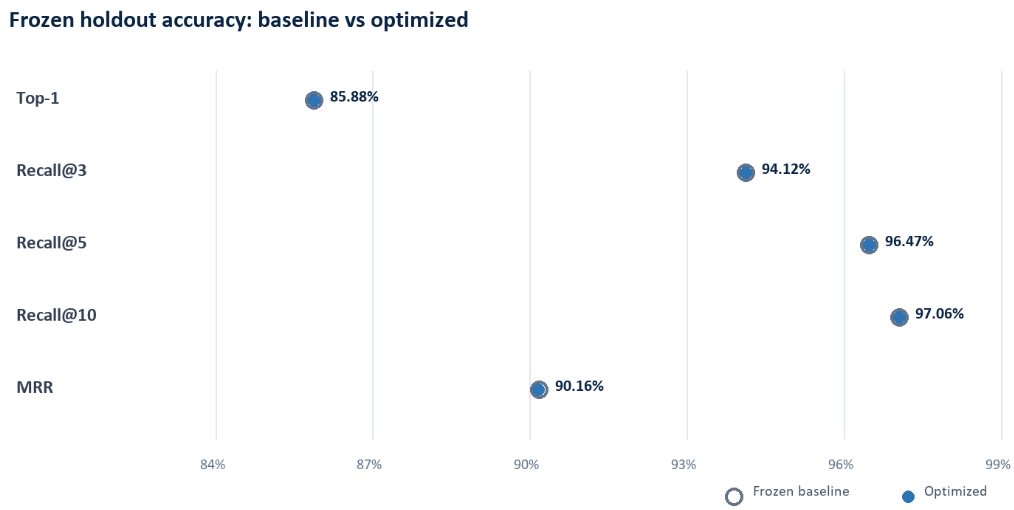

Figure 3. Frozen holdout ranking quality before and after optimization.

Accuracy metric	Frozen baseline	Optimized	Delta
Top-1 accuracy	85.88%	85.88%	0.00 pp
Recall@3	94.12%	94.12%	0.00 pp
Recall@5	96.47%	96.47%	0.00 pp
Recall@10	97.06%	97.06%	0.00 pp
Mean Reciprocal Rank	90.1816%	90.1599%	-0.0218 pp
Expert failures	Not applicable	0 of 170	None

The 0.0218-percentage-point MRR movement is immaterial and did not change Top-1 or any reported recall threshold. It reflects ordering below the primary decision boundary rather than a loss of correct activity identification.

Why accuracy was preserved

The production path still creates Semantic, Engineering, Tree and Rescheduler-v2 candidate lists, fuses their union with the same frozen LambdaMART MetaRank models, calibrates the ranking separately and applies deterministic validators plus risk policy. The changes optimize when and how repeated inputs are prepared; they do not change the learned model or authority boundaries.

Remaining latency and next step

Horizun/Microsoft Project document startup remains the largest fixed cost at roughly 28 seconds in the observed run. The next high-value improvement is to keep the schedule service warm and stream linked evidence to the UI in small batches, so useful results appear before the full batch completes.

Evidence basis: VEDA durable job timestamps for project ae3bb68952fb4edf; direct 165-record resolver benchmark; frozen holdout_v032 with 170 labeled cases and 13,436 schedule activities; regression suites for MetaRank integration, retrieval edge cases, execution lifecycle, field capture/actuals, P6 write guards and provider fallback.